Resistance-design evidence: original PDF pages

Yue 2017: original PDF pages 3, 4, 7

NEA LBE Handbook 2015: original PDF pages 934, 936

Zhao 2020: original PDF pages 7

the loop, the momentum equation under steady-state condition, integrated along the whole loop, can be written as:

$$\Delta P_{\mathrm{D}} = \Delta P_{\mathrm{fr}}, \tag{1}$$

where ΔP_{D} is the driving force for the natural circulation and ΔP_{fr} the total friction along the loop.

The driving force is caused by density difference of LBE between the hot and cold legs and is related to the temperature difference (thermal buoyancy). According to the energy conservation law, the temperature difference can be calculated with the input power of the electrical rod simulator (P_{th}), the specific heat capacity of the fluid (c_{p}) and the mass flow rate of LBE ($\dot{m}$). Thus, the driving force term can be written as:

$$\Delta P_{\mathrm{D}} = \rho_0 \beta g H P_{\mathrm{th}} / (\dot{m} c_{\mathrm{p}}), \tag{2}$$

where ρ_0 is the fluid density corresponding to the average temperature in the loop, β is the fluid isobaric thermal expansion coefficient, g is gravitational acceleration, and H is the effective elevation change between heating and cooling centers.

To evaluate the total pressure drop along the flow path, a total singular pressure drop coefficient is defined as:

$$K = \sum_i \left[\left(k + f \frac{L}{D_{\mathrm{H}}} \right)_i \frac{A_{\mathrm{pipe}}^2}{A_i^2} \right], \tag{3}$$

where k is the singular pressure drop coefficient in the ith branch of the loop; f is the Darcy–Weisbach friction factor in the ith branch of the loop; L, D_{H} and A_i are the length, hydraulic diameter and flow area of the ith branch of the loop, respectively; and A_{pipe} is the main cross section along the flow path. The total singular pressure drop coefficient is discussed again in Sect. 3.2.

The friction term in Eq. (1) can be expressed as:

$$\Delta P_{\mathrm{fr}} = K \frac{1}{2} \rho_0 v^2 = \frac{1}{2} K \frac{\dot{m}^2}{\rho_0 A_{\mathrm{pipe}}^2}, \tag{4}$$

where v is the velocity in main pipe.

Assuming that K does not depend on the velocity, and using Eqs. (1), (2) and (4), the mass flow rate in the natural circulation flow regime can be calculated by Eq. (5).

$$\dot{m} = \left(\frac{2 \beta \rho_0^2 g H A_{\mathrm{pipe}}^2 P_{\mathrm{th}}}{c_p K} \right)^{\frac{1}{3}} \tag{5}$$

The $\dot{m}$ and K can be optimized as a function of the Reynolds number. Finally, the results can be optimized by iteration.

3.2 Resistance coefficients of parts and components

The resistance coefficient of the loop in Eq. (3) can be obtained by a thermal-hydraulic modeling for different parts of the primary loop. The total singular pressure drop coefficient K can be expressed by:

$$K = K_{\mathrm{HS}} + K_{\mathrm{EV}} + K_{\mathrm{pipe}}, \tag{6}$$

where K_{HS} is the pressure drop coefficient in HS; K_{EV} is the pressure drop coefficient in expansion vessel; and K_{pipe} is the pressure drop coefficient along the pipe. Since the cross section of the tube in HX is coincident with the cross section of the pipe, K_{pipe} contains the pressure drop in HX, and the pressure drop in two 90° elbows is included in K_{pipe}, too.

As the flow area in the HS differs from the pipe cross section, and the flow direction changes, the pressure drop coefficient in HS can be written as:

$$K_{\mathrm{HS}} = K_{\mathrm{in}} + K_{\mathrm{out}} + K_{\mathrm{l,eff}}, \tag{7}$$

where K_{in} is the pressure drop coefficient of the HS inlet, consisting of the effects of changes in flow direction (a T-shaped elbow with one end blocked, $K = 1.29$) [21] and the flow area ($K \approx 0.64$); K_{out} is the pressure drop coefficient of the HS outlet, mainly influenced by the change in flow area; and $K_{\mathrm{l,eff}}$ is the pressure drop coefficient along the HS, defined as:

$$K_{\mathrm{l,eff}} = \frac{f L_{\mathrm{HS}}}{D_{\mathrm{H}}} \left(\frac{A_{\mathrm{pipe}}}{A_{\mathrm{HS}}} \right)^2, \tag{8}$$

where f is the Darcy–Weisbach friction factor; L_{HS} is the HS length; D_{H} is hydraulic diameter of the HS; and $A_{\mathrm{HS}} = A - A_{\mathrm{r}}$ is flow area of the HS, with A being the cross section of HS and A_{r}, cross section of the heater.

The pressure drop in expansion vessel is caused by the inlet ($K \approx 1$), the outlet ($K \approx 0.5$) and the change in flow direction, and some chemical control instrumentation ($K \approx 1$) as well. According to previous experience on other loops of KYLIN-II facility, the resistance coefficient of the expansion vessel is $K_{\mathrm{EV}} \approx 2.5$.

It is assumed conservatively that the total length of the pipe (I.D. = 26 mm) is $L_{\mathrm{pipe}} \approx 11$ m with 2°–90° elbows ($K = 0.5 \times 2 = 1$, the other two elbows for the expansion vessel and the HS, respectively) [21]. The resistance coefficient, for the pressure drop along the pipe and 2 (90°) elbows, can therefore be expressed as:

$$K_{\mathrm{pipe}} = \frac{f L_{\mathrm{pipe}}}{D} + 1. \tag{9}$$

3.3 The preliminary results of theoretical analysis

The thermal-hydraulic performance in natural circulation can be calculated with geometrical configuration of the loop known. According to the approaches in Sect. 3.2, the hydraulic parameters of Reynolds numbers, resistance coefficients and overall pressure loss can be calculated as a function of the LBE mass flow rate. The main results are listed in Table 2. The total singular pressure drop

coefficient declines with the mass flow rate, mainly because the K_{pipe} declines with increasing Reynolds number.

By solving Eq. (5), one can estimate the equilibrium mass flow rate of natural circulation in the loop at different input powers. The main thermal-hydraulic parameters at different power levels are summarized in Table 3.

Table 3 Thermal-hydraulic performance matrix in natural circulation

P (kW)	$\dot{m}$ (kg s^{-1})	ΔT (°C)	v (m s^{-1})
4	0.657	41.4	0.119
8	0.834	65.3	0.151
12	0.957	85.3	0.174
16	1.058	102.9	0.192
20	1.143	119.1	0.208
24	1.215	134.4	0.221

4 The CFD simulation

4.1 Description of CFD modeling

CFD simulations were carried to evaluate the KYLIN-II natural circulation loop, together with the expansion tank. The geometry model of expansion tank, heating rod and HX was adopted as shown in Fig. 2. The valve at the bottom was replaced by a tube. The 3D geometry was modeled in ICEM CFD (the Integrated Computer Engineering and Manufacturing code for Computational Fluid Dynamics) with dimensions of the pipes and other key component described in Sect. 2. The heating volume did not consist of solid volume; only a surface source on the pin wall was adopted. A commercial code FLUNET was employed to simulate the steady state of natural circulation at different heating powers.

The simulation model is RNG κ-ε model with standard wall treatment [11]. The heating powers of 4.3, 7.6, 11.2, 15.3 and 17.9 kW were decided on experimental basis. Correspondingly, the heat fluxes of 5.15, 9.20, 13.47, 18.40 and 21.52 W cm^{-2} were imposed on the rod bundle surface. Regarding the HX, a heat transfer coefficient of 4.89 W cm^{-2} K^{-1} and a free stream temperature of 200 °C were utilized on the oil side. After a sensitivity analysis on the unstructured mesh, the grid numbers were defined as 1.3×10^6 °C.

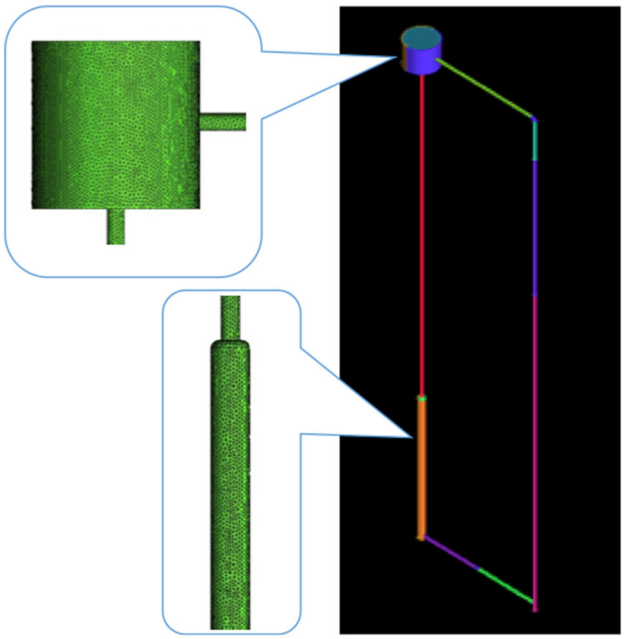


Fig. 2 3D geometry model and unstructured grid mesh of the loop

4.2 CFD simulation results of the steady-state experiments

The CFD results at different input powers were obtained using FLUENT. The temperature and velocity contours at heating power of 17.9 kW are shown in Fig. 3. The maximum temperature appeared on upper end of the heating rod (Fig. 3a). The loop was assumed adiabatic, without heat losses, so the outlet temperature of HX matches with the inlet temperature of HS. The HS outlet temperature was 334.4 °C, and the temperature difference through the HS was 101.5 °C. From Fig. 3b, the velocity through all pipes was almost constant, hence a steady state.

Table 2 Reynolds numbers, resistance coefficients and overall pressure loss as a function of the LBE mass flow rate

$\dot{m}$ (kg s^{-1})	Re_{tube} ($\times 10^3$)	Re_{HS} ($\times 10^3$)	K_{HS}	K_{EV}	K_{pipe}	K_{total}	ΔP_{fr} (Pa)
0.25	5.56	1.90	2.9	2.5	19.1	24.5	261.9
0.5	11.10	3.81	2.8	2.5	18.9	24.2	1034.8
0.75	16.70	5.71	2.8	2.5	18.5	23.8	2289.9
1	22.30	7.62	2.7	2.5	18.1	23.3	3985.4
1.25	27.80	9.52	2.6	2.5	17.7	22.8	6093.6

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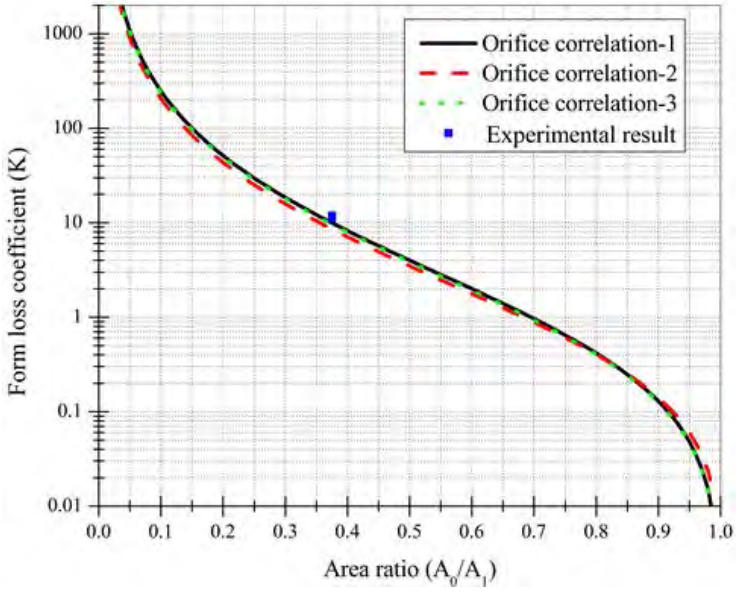
Table 15.3.3: Formulas for calculation of form loss coefficients in orifices

	Correlations	Applicable range	Reference
1	$K = \left(1 + 0.707\sqrt{1 - \frac{A_0}{A_1}} - \frac{A_0}{A_1}\right)^2 \left(\frac{A_1}{A_0}\right)^2$	$Re > 10^5$	Idelchik (1994)
2	$K = \left(1 - \frac{A_0}{A_1} + \sqrt{\xi}\left(1 - \frac{A_0}{A_1}\right)^{0.375}\right)^2 \left(\frac{A_1}{A_0}\right)^2$ where $\xi = 0.13 + 0.34 \times 10^{-\left[3.4(l/D) + 88.4(l/D)^{2.3}\right]}$	$Re > 10^4$	Idelchik (1994)
3	$K = \left(0.5\left(1 - \frac{A_0}{A_1}\right) + \left(1 - \frac{A_0}{A_1}\right)^2 + \tau\sqrt{1 - \frac{A_0}{A_1}}\left(1 - \frac{A_0}{A_1}\right)\right)\left(\frac{A_1}{A_0}\right)^2$ $\tau = (2.4 - \bar{l}) \times 10^{-\varphi(\bar{l})}$ $\varphi(\bar{l}) = 0.25 + \frac{0.535\bar{l}^{-8}}{0.05 + \bar{l}^{-8}}$	$Re > 10^3$	Idelchik (1994)

Where *f, r, D, Re_D* and ε are friction factor, radius of pipe, diameter of pipe, Reynolds number and absolute surface roughness, respectively.

All the correlations used in prediction of pressure loss for an orifice depend on the ratio of pipe flow area to orifice tap small area, regardless of Reynolds number, as shown in Figure 15.3.3. The area ratio (A_0/A_1) in the orifice of HELIOS is about 3.8. Experimental results (having a range of 10.9-11.6, almost constant independent of Reynolds numbers) show good agreement with used correlations.

Figure 15.3.3: Comparison of orifice correlations with experimental result



15.3.4 *Spacers*

Pressure losses of core or heat exchanger are dominant among those of the fluid system because of its small flow area and complex geometry. Grid spacers that support the core rods or steam generator tubes are the highest regions of pressure losses. Rehme (1973)

In contrast the detailed modelling of the complex geometry using CFD tools produced results that agree well with HELIOS experimental data, as shown in the figure above. It has been found that CFD modelling always provided more accurate results than experimental correlations (Cho, 2011). Further examples are given in Section 15.3.6.

15.3.5 *Sudden area change*

In regions like the gate valve and connecting piece of flange section, flow areas suddenly change, which disturbs the flow. The sudden area changes are illustrated in Figure 15.3.6 and correlations are listed in Table 15.3.5.

Figure 15.3.6: Geometries of sudden area changes

Sudden expansion (left), sudden contraction (right)

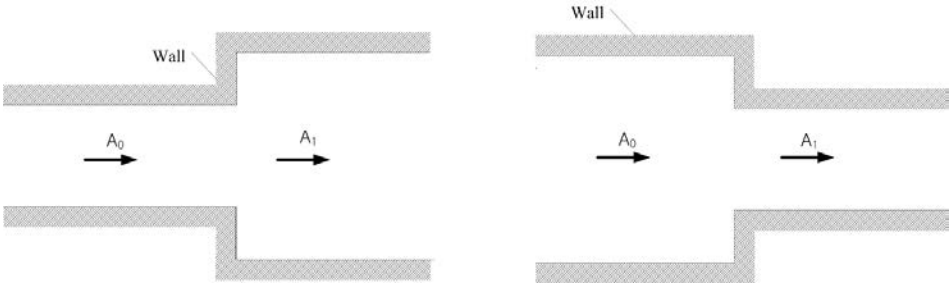


Table 15.3.5: Formulas for calculation of form loss coefficients in sudden expansion of flow area

Sudden expansion	Range	Reference
$K = \left(1 - \frac{A_0}{A_1}\right)^2$ Borda-Carnot's formula	$Re > 3.3 \cdot 10^3$	Idelchik (2000)

Table 15.3.6: Formulas for calculation of form loss coefficient in sudden contraction of flow area

	Sudden contraction	Range	Reference
1	$K = 0.5\left(1 - \frac{A_1}{A_0}\right)$	$Re > 10^4$	Idelchik (2000)
2	$K = 0.5\left(1 - \frac{A_1}{A_0}\right)^{3/4}$	$Re > 10^4$	Idelchik (1992)
3	$K = 0.5 - 0.7 \cdot \left(\frac{A_1}{A_0}\right) + 0.2 \cdot \left(\frac{A_1}{A_0}\right)^2$	$Re > 10^4$	Massey (1968)
4	$K = \left(1 - \frac{A_1}{A_c}\right)^2$ $\frac{A_c}{A_1} = 0.62 + 0.38\left(\frac{A_1}{A_0}\right)^3$	All range	Vennard (1961)

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